

AURCHI CHOWDHURY

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EDUCATION

Bangladesh University of Engineering and Technology

Nov 2022 – Present

Bachelor of Engineering in Computer Science and Engineering | CGPA: 3.88/4.00

Dhaka, Bangladesh

Viqarunnisa Noon School and College

Jan 2009 – Dec 2021

High School | GPA: 5.00/5.00 | Board Scholarship

Dhaka, Bangladesh

INTERESTS

Representation Learning · Natural Language Processing · Multimodal AI · Computer Vision · Mechanistic Interpretability

PUBLICATIONS

[🔗 Causal Localization of the English Pivot in LLaVA: Mechanistic VLM Analysis and Training-Free Multilingual Steering](#)

Jul 2026

1st Workshop on Multilinguality in the Era of Large Language Models (MeLLM), ACL 2026 | Archival

San Diego, CA

[📄 ACL Anthology](#)

[🔗 Where Do Affordances Crystallize? Video vs. Image Self-Supervised Encoders for Embodied Perception](#)

Jun 2026

7th Annual Embodied AI (EAI) Workshop, CVPR 2026 | Non-archival Poster

Denver, CO

[📄 EAI Workshop Paper](#)

WORK EXPERIENCE

Undergraduate Research Intern

On-site

[🔗 ATLAS: Adaptive Tutoring & Learning Assessment System](#) | Dept. of CSE, BUET

Nov 2025 – Sep 2026

- Developed an adaptive tutoring platform using Bayesian Knowledge Tracing (BKT), which builds learning pathways aligned with the Higher Secondary Certificate (HSC) exam syllabus.
- Supervisor: [Md. Shariful Islam Bhuiyan](#)
- [🔗 Code](#)

PROJECTS

[🔗 Bengali Long-Form ASR and Diarization](#) | *PyTorch, Whisper, Pyannote, Silero VAD*

Mar 2026

- Tackled low-resource Bengali **Natural Language Processing** by building **WhisperAlign**, a **cross-attention**-guided word-boundary-aware chunker for fine-tuning **Whisper-medium** on Bengali ASR, and domain-adapting **Pyannote**'s segmentation head with **Dual-VAD intersection** for speaker diarization.
- [🔗 Code](#) | [📄 Paper](#) | [📄 Slides](#)

[🔗 MemoryLane](#) | *PyTorch, ConvNeXt, U-Net, PatchGAN, VGG*

Apr 2026

- Approached grayscale image colorization as a **Computer Vision** problem of conditional image generation in CIELab color space, building a **ConvNeXt-Base encoder-U-Net decoder** with a **PatchGAN** discriminator. Reduced the dull, grayish outputs common in colorization models by training on colorfulness-filtered data with a combined **perceptual + adversarial loss** (VGG + PatchGAN).
- [🔗 Code](#) | [📄 Slides](#)

[🔗 Scramble](#) | *ATMega32, SG90 servo motors, HC-05 Bluetooth module, Kotlin*

Jul 2025

- Developed a Fully Automated Rubik's Cube Solver capable of generating solutions in a maximum of 20 moves.
- [🔗 Code](#) | [📺 Demo](#)

[🔗 On the Road](#) | *Node, Express, React, PostgreSQL*

Mar 2024

- Built a full-stack travel planning platform covering accommodations, dining, attractions, and tour guides.

ACHIEVEMENTS

- 🏆 11th / 124, **DL Sprint 4.0: Bengali Long-Form ASR & Diarization | BUET CSE Fest 2026** Mar 2026
7th / 124 in ASR and 4th / 124 in Diarization; built **WhisperAlign** for low-resource Bengali ASR and domain-adapted **Pyannote** with **Dual-VAD intersection** for speaker diarization.
- 🌐 1st Runner Up, **ResearchSpark 3 Minute Thesis Competition | BUET** Apr 2026
Built **BanglaHateCorpus** (168K instances) and **BanHateFormer**, a **Mixture-of-Experts** Transformer achieving 91.33% F1 at 94× smaller scale than BanglaBERT for **NLP-based content moderation**.
- 🌐 1st Runner Up, **HULT Prize at BUET** Feb 2026
Pitched **ATLAS**, a **multimodal AI** tutoring platform integrating **Vision Transformer OCR**, **Bayesian Knowledge Tracing**, and **BanglaBERT-based NLP** grading for 24M underserved students.
- 🏆 2nd Runners Up, **Research Poster Presentation | BUET CSE Fest 2024** Oct 2024
Designed a real-time **multimodal threat detection** system combining **NLP-based rumor classification**, **CNN+RNN** video analysis, and a confidence-scoring fusion module across 16 threat classes.